Theorem 1 (Max-Flow Min-Cut Theorem)**.** *Let $G = (V, E)$ be a directed network with a nonnegative capacity function $u : E \to \mathbb{R}_{\geq 0}$, together with a source s and a sink t. The value of an s–t flow is the total amount of flow leaving s. An s–t cut is a subset $S \subseteq V$ such that $s \in S$ and $t \notin S$, and its capacity is*

$$\mathrm{cap}(S) = \sum_{(i,j)\in E,\ i\in S,\ j\notin S} u(i,j).$$

Then

$$\max\{\textit{value of a feasible } s\text{–}t \textit{ flow}\} = \min\{\mathrm{cap}(S) : s \in S,\ t \notin S\}.$$

In other words, the maximum value of a feasible flow equals the minimum capacity of an s–t cut.

Proof. Let x be any feasible s–t flow, and let its value be z. For any s–t cut S, sum the flow-conservation equations over all vertices in $S \setminus \{s\}$. All contributions on edges with both endpoints in S cancel, and we obtain the cut identity

$$z = \sum_{(i,j)\in E,\ i\in S,\ j\notin S} x_{ij} - \sum_{(j,i)\in E,\ j\notin S,\ i\in S} x_{ji}.$$

Since $0 \leq x_{ij} \leq u(i,j)$ on every edge, it follows that

$$z \leq \sum_{(i,j)\in E,\ i\in S,\ j\notin S} u(i,j) = \mathrm{cap}(S).$$

Thus every feasible flow has value at most the capacity of every s–t cut.

For the reverse inequality, run the augmenting-path algorithm until it stops, and let x^* be the resulting flow. In the residual network of x^*, let S be the set of vertices reachable from s. Because there is no augmenting path, $t \notin S$, so S is an s–t cut.

Now consider an edge (i,j) with $i \in S$ and $j \notin S$. If it were not saturated, then the residual network would contain the forward edge $i \to j$, so j would also be reachable from s, a contradiction. Hence $x^*_{ij} = u(i,j)$ for every such edge.

Next consider an edge (j,i) with $j \notin S$ and $i \in S$. If $x^*_{ji} > 0$, then the residual network would contain the reverse edge $i \to j$, again making j reachable from s, a contradiction. Hence $x^*_{ji} = 0$ for every edge entering S from $V \setminus S$.

Substituting these two facts into the cut identity gives

$$\text{value}(x^*) = \sum_{(i,j)\in E,\ i\in S,\ j\notin S} u(i,j) = \text{cap}(S).$$

So there exists a feasible flow whose value equals the capacity of some s–t cut. Combined with the first part, this proves that the maximum flow value equals the minimum cut capacity. $\square$